Dell CloudLink 8.3.x

Upgrade Guide

June 2026
Rev. 1.0

DELL Technologies

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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Revision history

Table 1. Revisions

Date	Document revision	Description of changes
June 2026	1.0	Initial release

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About Dell CloudLink

This guide describes the process to upgrade to Dell CloudLink version 8.3.x. Because of the critical nature of the encryption keys and CloudLink Agent information that is contained in CloudLink Center servers, review the information in this document before starting the upgrade process.

CloudLink secures sensitive information in VMs across both public and private clouds. It provides encryption for the boot volume and additional data volumes with pre-startup authorization for cloud-hosted VMs. CloudLink provides this encryption by using native operating system encryption features: Microsoft BitLocker for Windows or dm-crypt for Linux.

CloudLink enables you to use native operating system encryption features to encrypt the boot and data volumes of the VM in a multitenant cloud environment. This encryption helps protect the integrity of the VM itself against unauthorized modifications. CloudLink encrypts the virtual machine boot and data volumes with unique keys that are controlled by enterprise security administrators. Cloud administrators or other tenants in the cloud do not have access to the keys. Securing VMs enables you to define the security policy it must meet before passing prestartup authorization, including verifying the integrity of the boot chain of the VM. This offers protection against tampering.

CloudLink ensures that only trusted and verified VMs can run and access sensitive data stored in the cloud. As part of the CloudLink solution, CloudLink Center defines the key release policy, performs prestartup authorization, and monitors the CloudLink Agents, events, and logs.

CloudLink also offers significant benefits for environments that use Dell PowerFlex resources. PowerFlex is a Software-Defined Solution (SDS) that enables you to transform Direct-Attached Storage (DAS) on existing hardware into shared block storage. It offers considerable scalability and extreme performance with flexible and elastic storage capacity and nodes. CloudLink encrypts the storage node devices with unique keys which the enterprise security administrators control.

CloudLink Center—The web-based management interface for CloudLink is used for managing encryption keys, configuring security policies, and monitoring the security and operation events and logs.

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CloudLink Center upgrade requirements

Before upgrading to CloudLink version 8.3.x, ensure that the system meets the requirements that are listed in this chapter.

Upgrade requirements

For any security vulnerabilities, Dell Technologies recommends that you upgrade to the latest version available. For more information about the latest versions, see the knowledge base article Cloudlink-security-support-policy-and-security-update-release-schedule-for-supported-versions.

- For an upgrade from 8.2.x to 8.3.x (build 8.3.1-1.1.1.1 for PowerFlex Manager 5.1), follow the Upgrade CloudLink 8.2 to CloudLink 8.3.x process in this document to complete the upgrade.
- For an upgrade from 8.1.2 to 8.3.x (build 8.3.1-1.1.1.1 for PowerFlex Manager 5.1), follow the Upgrade CloudLink 8.1.2 to CloudLink 8.3.x process in this document to complete the upgrade.
- For an upgrade from 8.1.1 to 8.3.x, follow the Upgrade CloudLink 8.1.1 to CloudLink 8.3.x process in this document to complete the upgrade.
- For an upgrade from 8.1.0 to 8.3.x, follow the Upgrade CloudLink 8.1 to CloudLink 8.3.x process in this document to complete the upgrade.
- Log in to the CloudLink Center to verify the CloudLink version and the build number. The version and build number is displayed in the **Information** box on the **CloudLink Center Home** panel.
- Need an active Internet connection to download the latest packages for upgrading Ubuntu Server on Microsoft Azure and AWS.
- Clear all the alarms before upgrading to the latest CloudLink version.
- If integration to AD/LDAP is using SSL (not recommended), contact Dell Technologies Support team to re-configure Windows Domain with LDAP over SSL, as the configuration gets disabled during an upgrade.

For information about upgrading from CloudLink 7.1.5 or later versions to CloudLink version 8.3, see KB 000216470 .

VMs

- Ensure that all CloudLink Center virtual machines have at least 6 GB of RAM, four vCPUs, and 64 GB disk space before you upgrade. Ensure that CloudLink cluster servers are online and synchronized with each other. All CloudLink virtual machines must be upgraded before they synchronize. Hosts may appear disconnected until all CloudLink center virtual machines are upgraded.

  NOTE: Ensure that the **Machine Groups** > **Modify** > **Modify Group** > **Machine Agent Upgrade** is set to *Auto*.

  NOTE: If a CloudLink Center VM does not have enough resources for upgrading, the following message is displayed: **System does not meet the minimum requirements**.

- Post upgrade, ensure that your CloudLink Center virtual appliance is running the 8.3 (build 8.3.1-1.1.1.1 for PowerFlex Manager 5.1).
- To automatically upgrade all CloudLink Agents, ensure that all machines registered with CloudLink Center are online and indicate as **Connected** in CloudLink Center. The `Machine Agent Upgrade` policy must be set to **Auto**.
- If the Machine Agent Upgrade policy is set to **Auto**, any CloudLink Agent that is not connected to CloudLink Center is automatically upgraded the next time CloudLink connects. If machines are online and connected during the upgrade, verify after the upgrade process. For more information, see Postupgrade verification.
- If the Machine Agent Upgrade policy is set to **Manual**, then all the machines in the respective machine group that uses Machine Agent Upgrade policy must be upgraded manually.
- Ensure that CloudLink Center cluster servers are online and synchronized with each other.

Backups

It is recommended that you back up CloudLink Center and the VMs before starting the upgrade process. Perform backups by using VM snapshots or traditional backup tools.

Ensure that the following critical backup requirements are met:

- Back up your CloudLink Center configuration data. For more information, see "Back up and restore CloudLink Center" in the *Dell CloudLink Administration Guide*.
- If you are using Microsoft Active Directory, Amazon S3 external storage, or an S3-compatible bucket, back up your encryption keys by using tools that are specific to those environments.

Keystore

If you are using an external keystore, ensure that the keystore is connected before starting the upgrade process.

CloudLink licenses

If you are upgrading to CloudLink 8.3.x from an earlier version, the license of the earlier version is retained and is displayed on the CloudLink interface. However, for 8.3.x and later, you do not require explicit licenses to access CloudLink.

Passwords

You require the following passwords:

- The CloudLink secadmin
- The CloudLink console
- One of the CloudLink vaults
- Passwords of the encrypted virtual machines

CloudLink clusters

The upgrade process upgrades all servers in a cluster. Before starting the cluster upgrade process, ensure the following:

- All servers are accessible.
- The upgrade ISO is uploaded to any server in the cluster.
- The ISO image file is uploaded to all servers.

Ports

Before starting the upgrade process, ensure that TCP ports 1194 and UDP port 1194 are open. For more information, see the section Software requirements for deploying CloudLink Center, in the *Dell CloudLink Deployment Guide*.

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CloudLink upgrade overview

Upgrading CloudLink requires uploading the upgrade ISO image file and initiating the upgrade.

All servers in a CloudLink Center cluster are automatically upgraded during the upgrade process. If the Machine Agent Upgrade policy is set to **Auto**, the connected CloudLink Agents are automatically upgraded. Any CloudLink Agent that is not connected is automatically upgraded the next time it connects to CloudLink Center.

CloudLink enables you to manually upgrade CloudLink Agents after you upgrade CloudLink Center or a CloudLink Center cluster, if the Machine Agent Upgrade policy is set to **Manual**. See Upgrade CloudLink Agents manually.

NOTE: The CloudLink version 8.3 release is intended only for PowerFlex deployments.

Group policy and CloudLink agent

If the group policy in an Active Directory environment handles the CloudLink agent installation, then the CloudLink Agent `Group Policy Object` must be updated with the new CloudLink Agent MSI file. To upgrade the policy object, do the following:

1. Delete the old CloudLink Agent MSI installer file from its Group Policy Object.
2. Upgrade CloudLink Center using the upgrade ISO image file.
3. Add the version 8.3.x CloudLink Agent MSI installer file to the `Group Policy Object` .

Change CloudLink Vault unlock modes

Use this procedure to change CloudLink Vault unlock modes.

Prerequisites

The CloudLink Vault must be set to **Auto Unlock** mode before upgrading the agents. This allows CloudLink Center to unlock immediately after the upgrade process begins so that all CloudLink Agents can be upgraded.

About this task

You can also set CloudLink Vault to **Manual Unlock** before the upgrade. But, you must:

- Enter one of the CloudLink Vault passcodes when prompted.
- Unlock CloudLink Center after the upgrade process.

NOTE: It is recommended to set the CloudLink Vault to **Auto Unlock** mode before starting the upgrade process.

Steps

1. Change the CloudLink Vault to the auto unlock mode:
   a. Log in to CloudLink Center.
   b. Click **System** > **Vault**.
   c. On the **Vault** page, click **Actions** > **Change Mode**.
   d. In the **Confirm Unlock Mode Change** dialog box, when prompted to confirm the auto unlock mode change request, click **Change**.
2. Change the CloudLink Vault to the manual unlock mode:
   a. Log in to CloudLink Center.
   b. Click **System** > **Vault**.
   c. On the **Vault** page, click **Actions** > **Change Mode**.
   d. In the **Confirm Unlock Mode Change** dialog box, when prompted with the confirm the manual unlock mode change request, click **Change**.

Upgrade CloudLink 8.3.x to CloudLink 8.3.x using backup and restore method

Use this procedure to upgrade from CloudLink version 8.3.x to CloudLink version 8.3.x using the backup and restore method.

Steps

1. Log in to CloudLink Center.
2. From the CloudLink Web console, generate a CloudLink backup.

   For more information about generating a CloudLink backup, see the section, *Back up and restore CloudLink Center*, in the *Dell CloudLink Administration Guide*.
3. Download the key file locally.

   For more information about downloading key files, see *Dell CloudLink Administration Guide*.
4. Download the backup file locally.
   - If you have a CloudLink 8.3 cluster, backup from any of the cluster members.
   - For more information about backup files, see *Dell CloudLink Administration Guide*.
5. Power **OFF** the CloudLink VM.
6. Deploy a new CloudLink 8.3 standalone node.
7. To have the same static IP assigned to CloudLink 8.3, before powering **ON** the CloudLink 8.3 VM for the first time, configure FLEX FACTORY settings to get the same static IP which was assigned to CloudLink 8.3.
   a. Go to the VM from the vCenter Inventory.
   b. On the **Configure** tab, expand **Settings,** and select **vApp options** .
   c. Under **OVF Environment transport**, select **VMware Tools**, and click **OK** .
   d. Under **OVF Settings**, the **View OVF Environment** displays in disabled state because the VM is in Powered OFF state.
   e. Under **Properties**, click **ADD** and create Key, Label and Type:

      ```
      cl.interface.0.ip.4.netmask
      cl.interface.0.ip.4.gateway
      cl.interface.0.ip.4.address
      cl.hostname (optional)
      ```
   f. Select **SET VALUE** and set available. For example: static IP address, gateway, or netmask. Snapshot is post settings.
8. Power **ON** the CloudLink 8.3 VM.
9. Once CloudLink VM boots, the IP is set as same Static IP and able to launch a Web console.
10. As part of the first-time configuration steps, choose **Restore from Backup** option.
11. Select the backup file version as CloudLink 8.3.
12. Perform **Restore** only on **Standalone Node**.
13. Now upload the key file and backup file and click **Restore**.
14. Wait for a couple of minutes until restore completes successfully.
15. Login and check the successful restore to the CloudLink 8.3 version.

    You may add up to three more CloudLink 8.3 VMs to the existing CloudLink 8.3 to form a cluster. All CloudLink virtual machines must be upgraded before all cluster members are in synchronized status. CloudLink Agents may be disconnected until CloudLink virtual machines are in synchronized status.

    NOTE: Ensure that the **Machine Groups** > **Modify** > **Modify Group** > **Machine Agent Upgrade** is set to *Auto*.
16. **NOTE:** For DHCP IP, use the below commands to connect the CloudLink Agent back to the Restored CloudLink.

    NOTE: Contact Dell Technologies Support team when you require to upgrade the CloudLink version to 8.3 with Red Hat Enterprise Linux 9.x.

    For Windows Agent VM use `svm /S` *<New CLC IP>* and for Linux Agent VM use `svm -S`*<New CLC IP>*.
17. For Static IP flex factory settings, the Agent virtual machines get Auto upgraded and connect back successfully.

Upgrade CloudLink 7.1.5 and later versions to CloudLink 8.3.x

Use this procedure to migrate from CloudLink version 7.1.5 and later to CloudLink version 8.3.x.

Steps

1. Log in to CloudLink Center.
2. From the CloudLink (7.1.5 and later) Web console, generate a CloudLink backup.

   For more information about generating a CloudLink backup, see the section, *Back up and restore CloudLink Center*, in the *Dell CloudLink Administration Guide*.
3. Download the key file locally.

   For more information about downloading key files, see *Dell CloudLink Administration Guide*.
4. Download the backup file locally.
   - If you have a CloudLink 7.1.5 or 7.1.6 or 7.1.7 or 7.1.8 or 7.1.9 cluster, backup from any cluster member.
   - For more information about backup files, see *Dell CloudLink Administration Guide*.
5. Power **OFF** the CloudLink VM.
6. Deploy a new CloudLink 8.3.x standalone node.
7. To have the same static IP assigned to CloudLink 8.3.x, before powering **ON** the CloudLink 8.3.x VM for the first time, configure FLEX FACTORY settings to get the same static IP which was assigned to CloudLink 7.1.5 or 7.1.6 or 7.1.7 or 7.1.8 or 7.1.9.
   a. Go to the VM from the vCenter Inventory.
   b. On the **Configure** tab, expand **Settings,** and select **vApp options** .
   c. Under **OVF Environment transport**, select **VMware Tools**, and click **OK** .
   d. Under **OVF Settings**, the **View OVF Environment** displays in disabled state because the VM is in Powered OFF state.
   e. Under **Properties**, click **ADD** and create Key, Label and Type:

      ```
      cl.interface.0.ip.4.netmask
      cl.interface.0.ip.4.gateway
      cl.interface.0.ip.4.address
      cl.hostname (optional)
      ```
   f. Select **SET VALUE** and set available. For example: static IP address, gateway, or netmask. Snapshot is post settings.
8. Power **ON** the CloudLink 8.3.x VM.
9. Once CloudLink virtual machine boots, the IP is set as same Static IP and able to launch the Web console.
10. As part of the first-time configuration steps, choose **Restore from Backup** option.
11. Select the backup file version as CloudLink 7.1.5+.
12. Perform **Restore** only on **Standalone Node**.
13. Now upload the key file and backup file and click **Restore**.
14. Wait for a couple of minutes until restore completes successfully.
15. Login and check the successful restore to the CloudLink 8.3.x version.

    You may add up to three more CloudLink 8.3.x VMs to the existing CloudLink 8.3.x to form a cluster. All CloudLink virtual machines must be upgraded before all cluster members are in synchronized status. CloudLink Agents may be disconnected until CloudLink virtual machines are in synchronized status.

    NOTE: Ensure that the **Machine Groups** > **Modify** > **Modify Group** > **Machine Agent Upgrade** is set to *Auto*.
16. **NOTE:** If DHCP IP exists, use the below commands to connect the CloudLink Agent back to the Restored CloudLink.

    NOTE: Contact Dell Technologies Support team when you require to upgrade the CloudLink version to 8.1.1 with Red Hat Enterprise Linux 9.x.

    For Windows Agent VM use `svm /S <New CLC IP>` and for Linux Agent VM use `svm -S<New CLC IP>`.
17. For Static IP flex factory settings, the Agent virtual machines get Auto upgraded and connect back successfully.

Post upgrade tasks

Follow and complete the post upgrade tasks after the upgrade .

Steps

1. In CloudLink Center, navigate to **System** > **Time** and re-configure the Time servers back to the original ones.
   The Time servers are replaced with default public ones during an upgrade.
2. In CloudLink Center, check if this alarm is triggered after the upgrade. If the `/dev/sda10 mounted /var file system full /reached threshold` alarm is triggered, open a service request to cleanup the file system.
3. **NOTE:** Contact Dell Technologies Support team when you require to migrate the CloudLink version to 8.1.1 with RH9.x.

   On vCenter, re-establish vCenter trust KMS to bring it to a healthy status in KMS key providers.

Upgrade CloudLink Agents manually

Use this task if you want to manually upgrade CloudLink Agents in a particular machine group. If you are using CloudLink Center for KMIP, then you do not have to perform the following procedure for upgrading agents.

Steps

1. Set the **Machine Agent Upgrade** policy to **Manual** for the machine group that contains machines that have CloudLink Agents you want to upgrade manually.
2. Upgrade CloudLink Center to version 8.3.x by performing steps 1 through 8 in Upgrade CloudLink 8.2 to CloudLink 8.3.
3. Log in to CloudLink Center.
4. Click **Agents** > **Machines**.
5. Select a machine in the list, and then click **Actions** > **Upgrade**. Repeat this step for every machine that requires a CloudLink Agent upgrade.

Upgrade CloudLink 8.2 to CloudLink 8.3.x

Use this procedure to upgrade CloudLink version 8.2 to CloudLink version 8.3.x.

Prerequisites

- Ensure that your account has the required **Update System** permission.
- Set CloudLink Vault to **Auto Unlock** mode before the upgrade begins. This setting enables CloudLink Center to unlock immediately after the upgrade so that all CloudLink Agents can be upgraded. If CloudLink Vault is set to the Manual Unlock mode before the upgrade, it remains in Manual Unlock mode after the upgrade and the vault is locked. You can change the vault back to the Manual Unlock mode after the upgrade. See Change CloudLink Vault unlock modes.

Steps

1. Ensure that the CloudLink Vault is set to the **Auto Unlock** mode.
2. Log in to CloudLink Center.
3. Click **System** > **Upgrade**.
4. Click **Upload**.
5. In the **Upload ISO** dialog box, click  to go to the upgrade ISO file, and then click **Upload**.
   If you are upgrading a CloudLink Center cluster, it can take several seconds for the ISO image file to upload to all nodes in a cluster. Do not attempt to upload the ISO file again during this time.
   NOTE: If the ISO image file ISO file is not automatically uploaded to all cluster nodes, manually upload the ISO file to each cluster node.
6. Click **Upgrade**.

7. In the **Confirm Host Upgrade** dialog box, click **Upgrade**.
   Your connection to CloudLink Center is lost during the upgrade process. After the upgrade operation is complete, the login page is displayed.
8. Log in to CloudLink Center again when the upgrade is complete and perform the Postupgrade verification.

   NOTE:

   If integration to AD/LDAP is using SSL (not recommended), Contact Dell Technologies Support team to reconfigure Windows Domain with LDAP over SSL, as the configuration gets disabled during an upgrade.

Upgrade CloudLink 8.1.2 to CloudLink 8.3.x

Use this procedure to upgrade CloudLink version 8.1.2 to CloudLink version 8.3.x.

Prerequisites

- Ensure that your account has the required **Update System** permission.
- Set CloudLink Vault to **Auto Unlock** mode before the upgrade begins. This setting enables CloudLink Center to unlock immediately after the upgrade so that all CloudLink Agents can be upgraded. If CloudLink Vault is set to the Manual Unlock mode before the upgrade, it remains in Manual Unlock mode after the upgrade and the vault is locked. You can change the vault back to the Manual Unlock mode after the upgrade. See Change CloudLink Vault unlock modes.

Steps

1. Ensure that the CloudLink Vault is set to the **Auto Unlock** mode.
2. Log in to CloudLink Center.
3. Click **System** > **Upgrade**.
4. Click **Upload**.
5. In the **Upload ISO** dialog box, click  to go to the upgrade ISO file, and then click **Upload**.
   If you are upgrading a CloudLink Center cluster, it can take several seconds for the ISO image file to upload to all nodes in a cluster. Do not attempt to upload the ISO file again during this time.

   NOTE: If the ISO image file ISO file is not automatically uploaded to all cluster nodes, manually upload the ISO file to each cluster node.
6. Click **Upgrade**.
7. In the **Confirm Host Upgrade** dialog box, click **Upgrade**.
   Your connection to CloudLink Center is lost during the upgrade process. After the upgrade operation is complete, the login page is displayed.
8. Log in to CloudLink Center again when the upgrade is complete and perform the Postupgrade verification.

   NOTE:

   If integration to AD/LDAP is using SSL (not recommended), Contact Dell Technologies Support team to reconfigure Windows Domain with LDAP over SSL, as the configuration gets disabled during an upgrade.

Upgrade CloudLink 8.1.1 to CloudLink 8.3.x

Use this procedure to upgrade CloudLink version 8.1.1 to CloudLink version 8.3.x.

Prerequisites

- Ensure that your account has the required **Update System** permission.
- Set CloudLink Vault to **Auto Unlock** mode before the upgrade begins. This setting enables CloudLink Center to unlock immediately after the upgrade so that all CloudLink Agents can be upgraded. If CloudLink Vault is set to the Manual Unlock mode before the upgrade, it remains in Manual Unlock mode after the upgrade and the vault is locked. You can change the vault back to the Manual Unlock mode after the upgrade. See Change CloudLink Vault unlock modes.

Steps

1. Ensure that the CloudLink Vault is set to the **Auto Unlock** mode.
2. Log in to CloudLink Center.
3. Click **System** > **Upgrade**.
4. Click **Upload**.
5. In the **Upload ISO** dialog box, click  to go to the upgrade ISO file, and then click **Upload**.
   If you are upgrading a CloudLink Center cluster, it can take several seconds for the ISO image file to upload to all nodes in a cluster. Do not attempt to upload the ISO file again during this time.
   NOTE: If the ISO image file ISO file is not automatically uploaded to all cluster nodes, manually upload the ISO file to each cluster node.
6. Click **Upgrade**.
7. In the **Confirm Host Upgrade** dialog box, click **Upgrade**.
   Your connection to CloudLink Center is lost during the upgrade process. After the upgrade operation is complete, the login page is displayed.
8. Log in to CloudLink Center again when the upgrade is complete and perform the Postupgrade verification.

   NOTE:

   If integration to AD/LDAP is using SSL (not recommended), Contact Dell Technologies Support team to reconfigure Windows Domain with LDAP over SSL, as the configuration gets disabled during an upgrade.

Upgrade CloudLink 8.1 to CloudLink 8.3.x

Use this procedure to upgrade CloudLink version 8.1 to CloudLink version 8.3.x.

Prerequisites

- Ensure that your account has the required **Update System** permission.
- Set CloudLink Vault to **Auto Unlock** mode before the upgrade begins. This setting enables CloudLink Center to unlock immediately after the upgrade so that all CloudLink Agents can be upgraded. If CloudLink Vault is set to the Manual Unlock mode before the upgrade, it remains in Manual Unlock mode after the upgrade and the vault is locked. You can change the vault back to the Manual Unlock mode after the upgrade. See Change CloudLink Vault unlock modes.

Steps

1. Ensure that the CloudLink Vault is set to the **Auto Unlock** mode.
2. Log in to CloudLink Center.
3. Click **System** > **Upgrade**.
4. Click **Upload**.
5. In the **Upload ISO** dialog box, click  to go to the upgrade ISO file, and then click **Upload**.
   If you are upgrading a CloudLink Center cluster, it can take several seconds for the ISO image file to upload to all nodes in a cluster. Do not attempt to upload the ISO file again during this time.
   NOTE: If the ISO image file ISO file is not automatically uploaded to all cluster nodes, manually upload the ISO file to each cluster node.
6. Click **Upgrade**.
7. In the **Confirm Host Upgrade** dialog box, click **Upgrade**.
   Your connection to CloudLink Center is lost during the upgrade process. After the upgrade operation is complete, the login page is displayed.
8. Log in to CloudLink Center again when the upgrade is complete and perform the Postupgrade verification.

   NOTE:

   If integration to AD/LDAP is using SSL (not recommended), Contact Dell Technologies Support team to reconfigure Windows Domain with LDAP over SSL, as the configuration gets disabled during an upgrade.

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Verify connection between VM and CloudLink Center after upgrade

Use this task to verify the connection between VM and CloudLink Center after upgrade.

Prerequisites

Ensure that CloudLink 8.3.x is successfully upgraded.

Steps

1. Ensure that all VMs are connected and stable in CloudLink Center.
   - CloudLink Agents on connected machines attempt to connect to CloudLink Center after the upgrade is complete.
   - If a machine is in a machine group that uses the **Auto** upgrade policy, the CloudLink Agent that is installed on that machine is automatically upgraded to version 8.3.x.
   - On all Windows VMs, run the `svm /S <CloudLink IP>` command to connect the CloudLink agent and CloudLink Center.
   - On all Linux VMs:
     a. Run the `svm -S <CLoudLink IP>` command to connect the CloudLink agent and CloudLink Center.
     b. Run the `svm about` command to ensure that the CloudLink Agent version appears as "Version 8.3.xxxx".
     c. Run the `svm status` command to ensure that the volume encryption type is listed as **dmcrypt**, **ecryptfs**, or **LUKS** volume types in the third column. Those volumes are legacy Linux formats used by CloudLink version 5.0 and earlier, and are no longer supported. It is recommended that you contact Dell Technologies Support team to decrypt the **ecryptfs** or **LUKS** volumes and encrypt them using **dmcrypt**.

     NOTE: Key rotation is not available for **ecryptfs** volumes until they are decrypted and reencrypted.

2. Ensure that all keystores are accessible by CloudLink Center.
3. In CloudLink Center, check if these alarms were triggered after the upgrade:
   - If the `Backup file wasn't downloaded/uploaded to store recently` alarm was triggered, download a backup of the new CloudLink 8.3.x.
   - If the `Vault unlock codes are not configured` alarm was triggered, set the CloudLink Vault passcodes as described in the *Dell CloudLink Administration Guide*.
   - If `The keystore is not accessible` alarm was triggered, check the alarm for detailed information. This alarm is cleared when the keystore locations are accessible.
4. If you changed the CloudLink Vault unlock mode from **Manual Unlock** to **Auto Unlock** before the upgrade, change it back to **Manual Unlock**. For more information, see Change CloudLink Vault unlock modes.
5. Restart the VMs to ensure that their boot processes are unaffected by the upgrade.

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Troubleshooting and getting help

Go to Dell Technologies Online Support and click **MyService360**. There are several options for contacting the Dell Technologies Technical Support team. To open a service request, you must have a valid support agreement. Contact your Dell Technologies Sales representative for details about getting a valid support agreement or with questions about your account.

Dell Technologies support, product, and licensing information is also available with your Dell Technologies account manager.